Wind Energy

Chapter 5

Big Question

How do people use wind as a source of energy?

The sun heats the sand, trees, soil, and shallow waters around a tropical island. In the afternoon, the air over the island is very warm and rises higher into the atmosphere. This causes cooler air from over the sea to blow in. The moving air produces waves on the water. Boats that had been using their diesel engines are able to hoist their sails and use a free source of energy. This is wind energy.

Near the island stands a series of tall wind turbines. The blades of the turbines rotate faster as the wind picks up. Inside each turbine, a generator produces electricity with each rotation of the blades. The electricity transfers through wires to a small power station. The station shuts down its noisy diesel generator. For the afternoon, the entire island is powered by the wind.

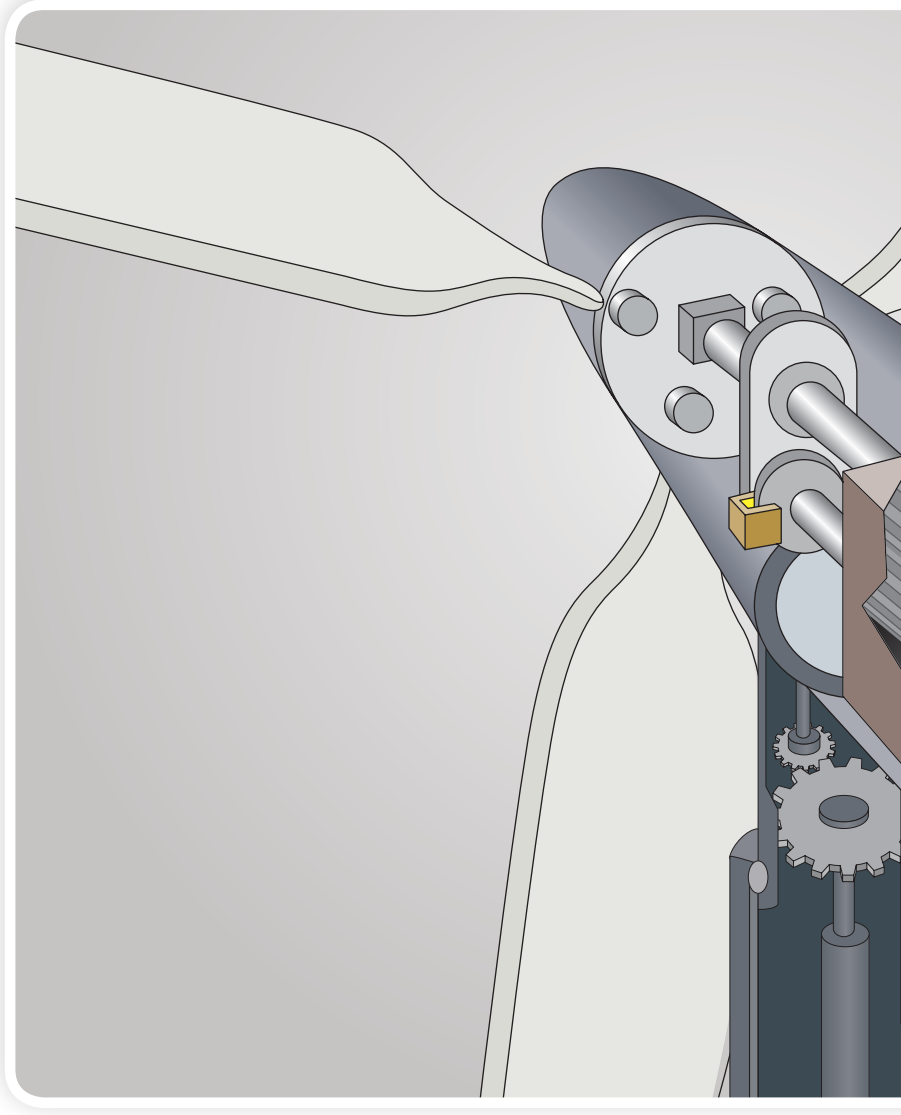


Moving air, or wind, can move a sailboat. It can also spin a turbine, allowing wind energy to become electricity.

Converting Wind Energy to Electricity

For centuries, windmills converted wind energy to mechanical energy. The blades turned a shaft that was connected to gears. These gears helped operate a millstone for grinding grain into flour.

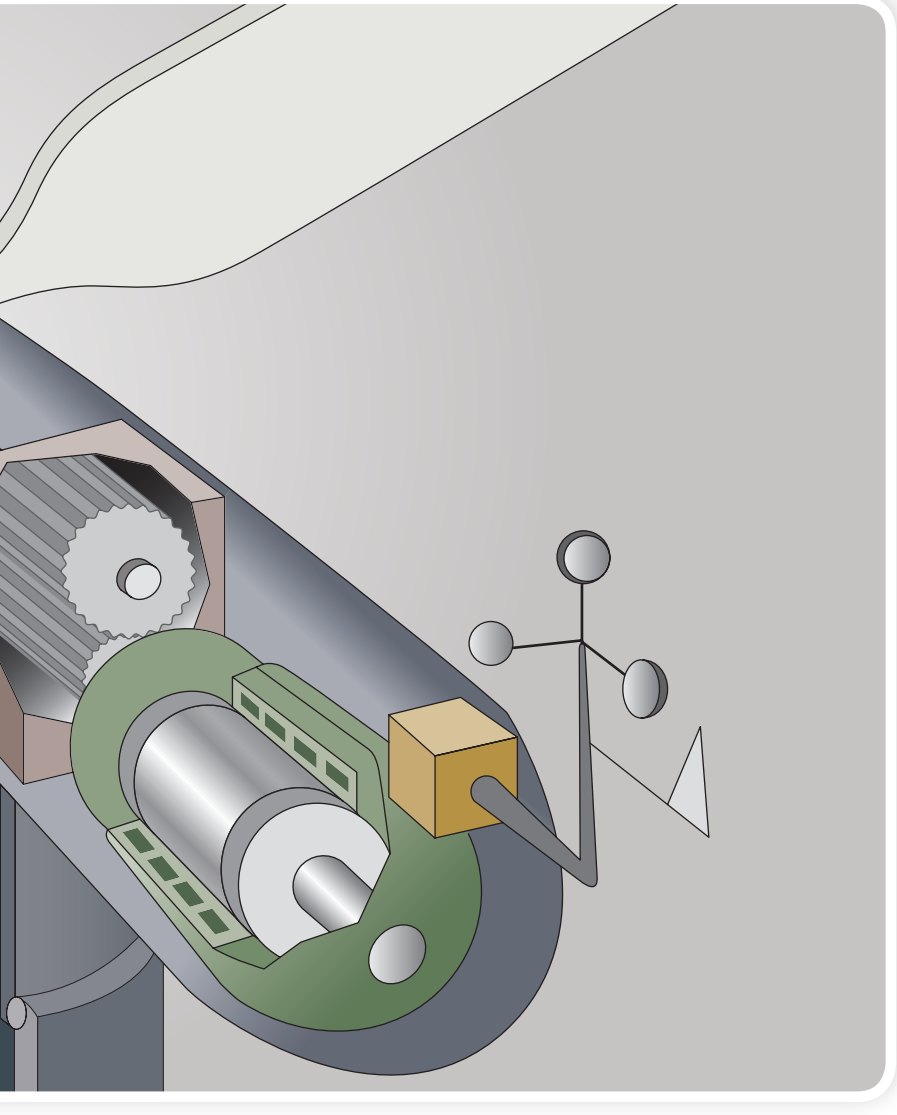
Modern wind turbines that convert wind energy into electricity have the added step of converting mechanical energy into electrical energy. In a modern wind turbine, the blades are attached to a large shaft with a large gear housed inside the turbine. The large



The relatively slow speed of the large shaft in the turbine is converted to high speed to generate electricity. This technology is made of heavy parts, so turbine towers need to be very strong.

gear rotates and is engaged with a smaller gear. The smaller gear is attached to another shaft inside a generator. The shaft rotates between 1,000 and 1,800 times per minute.

The direction that turbine blades face can be changed so the blades capture more or less wind. This allows engineers to prevent blades from spinning too fast when the wind is blowing too hard. Or it can prevent the blades from spinning at all when the wind is too weak to generate power.



Things to Consider About Wind Power

Wind is an abundant renewable resource, but it is not always reliable. Some locations can be windy one day and calm the next. Some have winds that blow for weeks and then periods of calm. Others have breezes that are steady but not strong enough to turn large turbine blades fast enough to generate electricity.

A **wind farm** consists of multiple turbines placed together. Wind farms can cover vast areas. Turbines need to be spaced far enough apart that each

Vocabulary

wind farm, n. many wind turbines installed together in an area to generate electricity for a community



one receives wind without disrupting how the others operate. Some wind power projects have been controversial because some people do not like the appearance of the turbines across the landscape.

Another factor that must be considered is how the turbines will be maintained and monitored. A turbine anchored to the ocean floor ten miles from the coast may be more difficult or expensive to maintain and monitor than a turbine on land. On the other hand, the offshore turbine may produce more power.



If you lived on the shore or even just went to the beach, would you like seeing this offshore wind farm on the water? Or would it bother you? If you were a fishing boat captain, would you appreciate seeing a renewable energy project in the waters you fish, or would you mostly be worried about having to navigate around it?

Wind Energy Transmission and Storage

Wind energy harnessed by a turbine becomes electricity. Transmission, or the movement, of electricity usually involves power lines that run from a turbine to the nearest electrical power grid. The grid is the network of power lines, towers, underground cables, transformers, and other technology that distributes electricity to homes, schools, and businesses. When a wind farm generates electricity, the people who operate the grid can direct the electricity to specific parts of the grid and keep track of how much is being used.

Electricity can also be stored in batteries. Because wind power is dependent on wind, which is not always blowing, batteries can help make wind power more reliable. Individual homes or neighborhoods can even have their own small wind farms. A large battery can store the electricity for later use. Wind power can be combined with other electricity sources.



In the future, renewable energy use might look more commonly like this—homes with solar panels on their roofs and a wind farm nearby.

Word to Know

A *power grid* is a network of power plants, transfer stations, and power lines that supplies electricity within a region.